

## Chapter 5: Correlation

### EXERCISE 5.1 [PAGE 63]

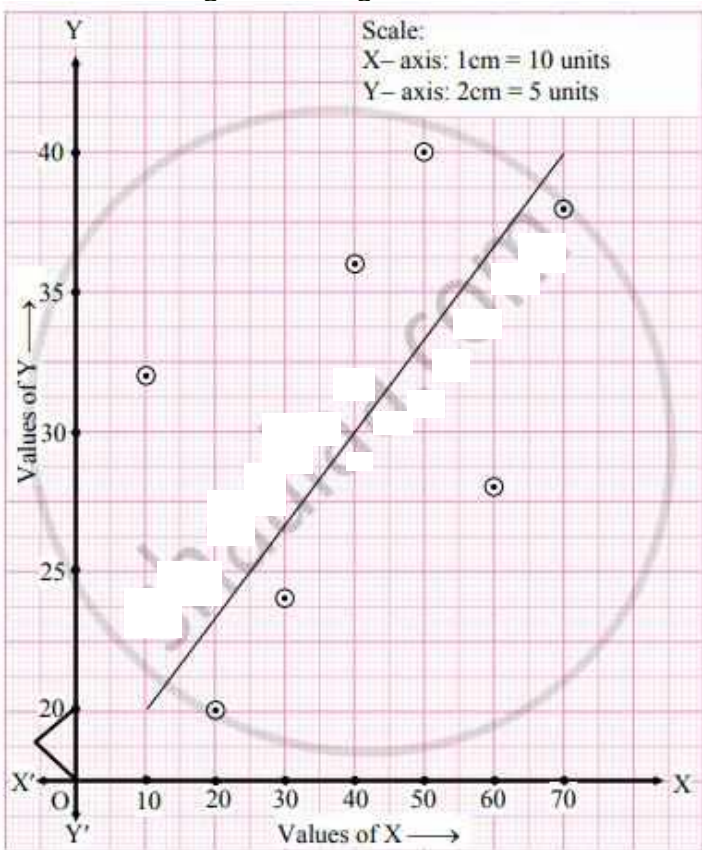
#### Exercise 5.1 | Q 1 | Page 63

Draw scatter diagram for the data given below and interpret it.

|   |    |    |    |    |    |    |    |
|---|----|----|----|----|----|----|----|
| x | 10 | 20 | 30 | 40 | 50 | 60 | 70 |
| y | 32 | 20 | 24 | 36 | 40 | 28 | 38 |

#### SOLUTION

The scatter diagram is as given below:



Since all the points lie in a band rising from left to right. Therefore, there is a positive correlation between the values of X and Y respectively.

#### Exercise 5.1 | Q 2 | Page 63

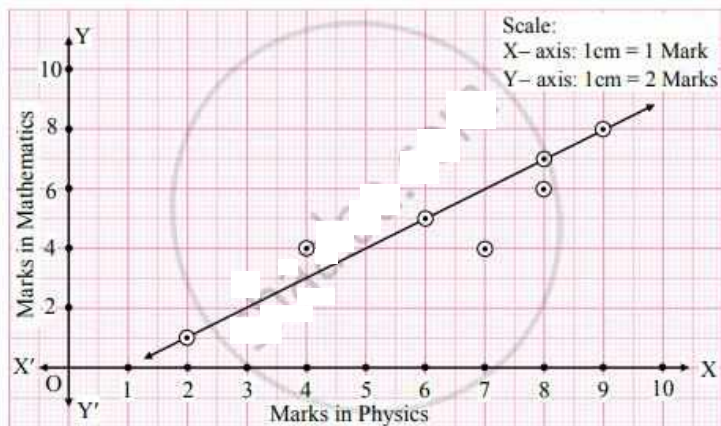
For the following data of marks of 7 students in physics (x) and Mathematics (y), draw scatter diagram and state the type of correlation.

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| x | 8 | 6 | 2 | 4 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| y | 6 | 5 | 1 | 4 | 4 | 7 | 8 |
|---|---|---|---|---|---|---|---|

### SOLUTION

We take marks in Physics on X-axis and marks in Mathematics on Y-axis and plot the points as below.



We get a band of points rising from left to right. This indicates the positive correlation between marks in Physics and marks in Mathematics.

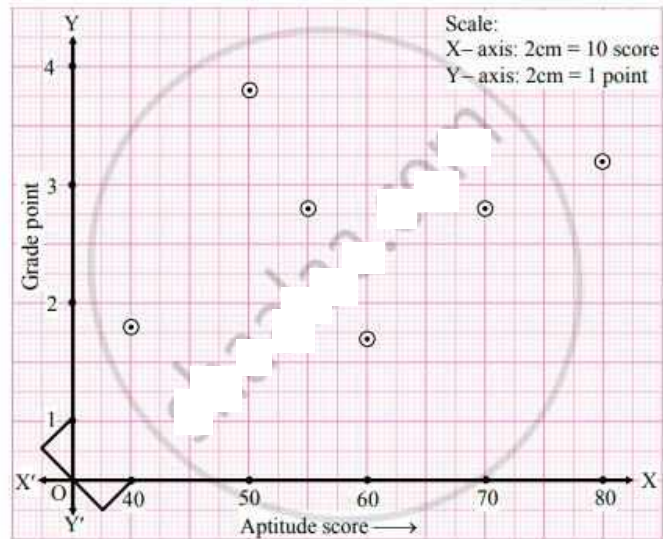
### Exercise 5.1 | Q 3 | Page 63

Draw scatter diagram for the data given below. Is there any correlation between Aptitude score and Grade points?

|                       |     |     |     |     |     |     |
|-----------------------|-----|-----|-----|-----|-----|-----|
| <b>Aptitude score</b> | 40  | 50  | 55  | 60  | 70  | 80  |
| <b>Grade points</b>   | 1.8 | 3.5 | 2.8 | 1.7 | 2.8 | 3.2 |

### SOLUTION

|                          |     |     |     |     |     |     |
|--------------------------|-----|-----|-----|-----|-----|-----|
| <b>Aptitude score(x)</b> | 40  | 50  | 55  | 60  | 70  | 80  |
| <b>Grade points (y)</b>  | 1.8 | 3.5 | 2.8 | 1.7 | 2.8 | 3.2 |



The points are completely scattered i.e., no trend is observed.

∴ There is no correlation between the Aptitude score (X) and Grade point (Y).

#### Exercise 5.1 | Q 4 | Page 63

Find correlation coefficient between x and y series for the following data.

$$n = 15, \bar{x} = 25, \bar{y} = 18, \sigma_x = 3.01, \sigma_y = 3.03, \sum (x_i - \bar{x})(y_i - \bar{y}) = 122$$

#### SOLUTION

here,  $n = 15$ ,  $\bar{x} = 25$ ,  $\bar{y} = 18$ ,  $\sigma_x = 3.01$ ,  $\sigma_y = 3.03$ , and  $\sum (x_i - \bar{x})(y_i - \bar{y}) = 122$

$$\text{Since, Cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

$$\therefore \text{Cov}(x, y) = \frac{1}{15} \times 122$$

$$= 8.13$$

$$\text{Since, } r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$\therefore = \frac{8.13}{3.01 \times 3.03}$$

$$= \frac{8.13}{9.1203}$$

$$\therefore r = 0.89$$

**Exercise 5.1 | Q 5 | Page 63**

The correlation coefficient between two variables x and y are 0.48. The covariance is 36 and the variance of x is 16. Find the standard deviation of y.

**SOLUTION**

Given,  $r = 0.48$ ,  $\text{Cov}(x, y) = 36$

Since  $\sigma_x^2 = 16$

$$\therefore \sigma_x = 4$$

$$\text{Since, } r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$\therefore 0.48 = \frac{36}{4 \times \sigma_y}$$

$$\therefore \sigma_y = \frac{36}{4 \times 0.48} = \frac{9}{0.48}$$

$$= \frac{900}{48} = 18.75$$

$\therefore$  Standard deviation of y is 18.75.

**Exercise 5.1 | Q 6. (i) | Page 63**

In the following data one of the value of y is missing. Arithmetic means of x and y series are 6 and 8 respectively.

$$(\sqrt{2} = 1.4142)$$

|          |   |    |    |   |   |
|----------|---|----|----|---|---|
| <b>x</b> | 6 | 2  | 10 | 4 | 8 |
| <b>y</b> | 9 | 11 | ?  | 8 | 7 |

Estimate missing observation

**SOLUTION**

Let the missing observation be 'a'

|          |   |    |    |   |   |
|----------|---|----|----|---|---|
| <b>x</b> | 6 | 2  | 10 | 4 | 8 |
| <b>y</b> | 9 | 11 | a  | 8 | 7 |

$$\bar{x} = 6, \bar{y} = 8, n = 5$$

$$\bar{y} = \frac{\sum y}{n} = \frac{35 + a}{5}$$

$$\therefore 8 = \frac{35 + a}{5}$$

$$\therefore a + 35 = 40$$

$$\therefore a = 40 - 35$$

$$\therefore a = 5$$

$$\therefore \text{Missing frequency} = 5$$

### Exercise 5.1 | Q 6. (ii) | Page 63

In the following data one of the value y of is missing. Arithmetic means of x and y series are 6 and 8 respectively.

$$(\sqrt{2} = 1.4142)$$

|          |   |    |    |   |   |
|----------|---|----|----|---|---|
| <b>x</b> | 6 | 2  | 10 | 4 | 8 |
| <b>y</b> | 9 | 11 | ?  | 8 | 7 |

Calculate the correlation coefficient

### **SOLUTION**

The missing observation be 'a'.

given,  $\bar{x} = 6, \bar{y} = 8, n = 5$

$$\bar{y} = \frac{\sum y}{n}$$

$$\therefore 8 = \frac{35 + a}{5}$$

$$\therefore 40 = 35 + a$$

$$\therefore a = 5$$

We construct the following table:

| x  | y  | x <sup>2</sup> | y <sup>2</sup> | xy  |
|----|----|----------------|----------------|-----|
| 6  | 9  | 36             | 81             | 54  |
| 2  | 11 | 4              | 121            | 22  |
| 10 | 5  | 100            | 25             | 50  |
| 4  | 8  | 16             | 64             | 32  |
| 8  | 7  | 64             | 49             | 56  |
| 30 | 40 | 220            | 340            | 241 |

Here,  $\sum x = 30$ ,  $\sum y = 40$

$$\sum x^2 = 220; \sum y^2 = 320$$

$$\sum xy = 214 \quad n = 5$$

∴ Karl Pearson's coefficient of correlation,

$$\begin{aligned}
 r &= \frac{n \sum xy - \sum x \sum y}{\sqrt{n \sum x^2 - (\sum x)^2} \sqrt{n \sum y^2 - (\sum y)^2}} \\
 &= \frac{5(214) - 30(40)}{\sqrt{5(220) - (30)^2} \sqrt{5(340) - (40)^2}} \\
 &= \frac{-130}{\sqrt{200} \sqrt{100}}
 \end{aligned}$$

$$\therefore = -0.919$$

∴ There is high degree negative correlation.

#### Exercise 5.1 | Q 7 | Page 63

Find correlation coefficient from the following data, [Given :  $\sqrt{3} = 1.732$ ]

|          |   |   |   |   |   |
|----------|---|---|---|---|---|
| <b>x</b> | 3 | 6 | 2 | 9 | 5 |
| <b>y</b> | 4 | 5 | 8 | 6 | 7 |

**SOLUTION**

| x         | i         | $x_i^2$    | $y_i^2$    | $x_i y_i$  |
|-----------|-----------|------------|------------|------------|
| 3         | 4         | 9          | 16         | 12         |
| 6         | 5         | 36         | 25         | 30         |
| 2         | 8         | 4          | 64         | 16         |
| 9         | 6         | 81         | 36         | 54         |
| 5         | 7         | 25         | 49         | 35         |
| <b>25</b> | <b>30</b> | <b>155</b> | <b>190</b> | <b>147</b> |

From the table, we have

$$n = 5, \sum x_i = 25, \sum y_i = 30, \sum x_i^2 = 155, \sum y_i^2 = 190, \sum x_i y_i = 147$$

$$\bar{x} = \frac{\sum x_i}{n}$$

$$= \frac{25}{5}$$

$$= 5$$

$$\bar{y} = \frac{\sum y_i}{n}$$

$$= \frac{30}{5}$$

$$= 6$$

$$\text{Since, Cov (x, y) = } \frac{1}{n} \sum x_i y_i - \bar{x} \bar{y}$$

$$\therefore \text{Cov}(x, y) = \frac{1}{5} \times 147 - (\bar{x})$$

$$= 29.4 - 30$$

$$= -0.6$$

$$\sigma_x^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$= \frac{155}{5} - (5)^2$$

$$= 31 - 25$$

$$\therefore \sigma_x^2 = 6$$

$$\therefore \sigma_x = \sqrt{6}$$

$$\sigma_y^2 = \frac{\sum y_i^2}{n} - (\bar{y})^2$$

$$= \frac{190}{5} - (6)^2$$

$$= 38 - 36$$

$$\therefore \sigma_y^2 = 2$$

$$\therefore \sigma_y = \sqrt{2}$$

$$\therefore \sigma_x \sigma_y = \sqrt{6} \sqrt{2} = \sqrt{12}$$

$$= 2\sqrt{3}$$

$$= 2(1.732) = 3.464$$

Thus, the correlation coefficient between x and y is

$$r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$= \frac{-0.6}{3.464}$$

$$= -0.1732$$

### Exercise 5.1 | Q 8 | Page 63

Correlation coefficient between x and y is 0.3 and their covariance is 12. The variance of x is 9, Find the standard deviation of y.

#### **SOLUTION**

Given:  $r = 0.3$ ,  $\text{Cov}(x, y) = 12$ ,

$$\sigma_x^2 = 9$$

$$\therefore \sigma_x = 3$$

Karl Pearson's coefficient of correlation,

$$r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$\therefore 0.3 = \frac{12}{3 \times \sigma_y}$$

$$\therefore \sigma_y = \frac{12}{3 \times 0.3}$$

$$\therefore \sigma_y = \frac{4}{0.3}$$

$$\therefore \sigma_y = 13.33$$

$\therefore$  The standard deviation of y is 13.33.

#### **MISCELLANEOUS EXERCISE 5 [PAGES 63 - 64]**

### Miscellaneous Exercise 5 | Q 1 | Page 63

Two series of x and y with 50 items each have standard deviations of 4.8 and 3.5 respectively. If the sum of products of deviations of x and y series from respective arithmetic means is 420, then find the correlation coefficient between x and y.

**SOLUTION**

Given,  $n = 50$ ,  $\sigma_x = 4.8$ ,  $\sigma_y = 3.5$ ,  $\sum (x_i - \bar{x})(y_i - \bar{y}) = 420$

$$\text{Cov}(x, y) = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$= \frac{1}{50} \times 420$$

$$\therefore \text{Cov}(x, y) = 8.4$$

$$r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$= \frac{8.4}{(4.8)(3.5)}$$

$$= \frac{84 \times 10}{48 \times 35}$$

$$= \frac{1}{2}$$

$$= 0.5$$

**Miscellaneous Exercise 5 | Q 2 | Page 63**

Find the number of pairs of observations from the following data,

$$r = 0.15, \sigma_y = 4, \sum (x_i - \bar{x})(y_i - \bar{y}) = 12, \sum (x_i - \bar{x})^2 = 40.$$

**SOLUTION**

Given,  $r = 0.15$ ,  $\sigma_y = 4$ ,  $\sum (x_i - \bar{x})(y_i - \bar{y}) = 12$ ,  $\sum (x_i - \bar{x})^2 = 40$

$$\text{Since, } \sigma_x = \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2} = \sqrt{\frac{40}{n}}$$

$$\text{Cov}(x, y) = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$= \frac{1}{n} \times 12$$

$$\therefore \text{Cov}(x, y) = \frac{12}{n}$$

$$\text{Since, } r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$\therefore 0.15 = \frac{\frac{12}{n}}{\sqrt{\frac{40}{n}} \times 4}$$

$$\therefore 0.15 = \frac{3}{n \times \sqrt{\frac{40}{n}}}$$

$$\therefore 0.15 = \frac{1}{\sqrt{n} \times \sqrt{40}}$$

Squaring on both the sides, we get

$$0.0025 = \frac{1}{n \times 40}$$

$$\therefore n = \frac{1}{0.0025 \times 40}$$

$$= \frac{10000}{25 \times 40}$$

$$= \frac{10000}{1000}$$

$$\therefore n = 10$$

Given that  $r = 0.4$ ,  $\sigma_y = 3$ ,  $\sum (x_i - \bar{x})(y_i - \bar{y}) = 108$ ,  $\sum (x_i - \bar{x})^2 = 900$ . Find the number of pairs of observations.

**SOLUTION 1**

Given,  $r = 0.4$ ,  $\sigma_y = 3$ ,  $\sum (x_i - \bar{x})(y_i - \bar{y}) = 108$ ,  $\sum (x_i - \bar{x})^2 = 900$

$$\text{Cov}(X, Y) = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$= \frac{1}{n} \times 108$$

$$\therefore \text{Cov}(X, Y) = \frac{108}{n}$$

$$\sigma_x = \sqrt{\frac{1}{n} \times \sum (x_i - \bar{x})^2}$$

$$= \sqrt{\frac{1}{n} \times 900}$$

$$= \sqrt{\frac{900}{n}} = \frac{30}{\sqrt{n}}$$

$$\text{Since, } r = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$$

$$\therefore 0.4 = \frac{\frac{108}{n}}{\frac{30}{\sqrt{n}} \times 3}$$

$$\therefore 0.4 = \frac{108}{n} \times \frac{\sqrt{n}}{30 \times 3}$$

$$\therefore 0.4 = \frac{12}{10\sqrt{n}}$$

$$\therefore \sqrt{n} = \frac{12}{4} = 3$$

$$\therefore n = 9.$$

**SOLUTION 2**

Given,  $r = 0.4$ ,  $\sigma_y = 3$ ,  $\sum (x_i - \bar{x})(y_i - \bar{y}) = 108$ ,  $\sum (x_i - \bar{x})^2 = 900$ .

$$\text{Cov}(x, y) = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$= \frac{1}{n} \times 108$$

$$\therefore \text{Cov}(x, y) = \frac{108}{n}$$

$$\sigma_x = \sqrt{\frac{1}{n} \times \sum (x_i - \bar{x})^2}$$

$$= \sqrt{\frac{1}{n} \times 900}$$

$$= \sqrt{\frac{900}{n}} = \frac{30}{\sqrt{n}}$$

$$\text{Since, } r = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

$$\therefore 0.4 = \frac{\frac{108}{n}}{\frac{30}{\sqrt{n}} \times 3}$$

$$\therefore 0.4 = \frac{108}{n} \times \frac{\sqrt{n}}{30 \times 3}$$

$$\therefore 0.4 = \frac{12}{10\sqrt{n}}$$

$$\therefore \sqrt{n} = \frac{12}{4} = 3$$

$$\therefore n = 9$$

**Miscellaneous Exercise 5 | Q 4 | Page 64**

Given the following information,  $\sum x_i^2 = 90$ ,  $\sum x_i y_i = 60$ ,  $r = 0.8$ ,  $\sigma_y = 2.5$ ,

where  $x_i$  and  $y_i$  are the deviations from their respective means, find the number of items.

**SOLUTION**

Here,  $r = 0.8$ ,  $\sum x_i^2 = 90$ ,  $\sum x_i y_i = 60$ ,  $\sigma_y = 2.5$ ,

Here,  $x_i$  and  $y_i$  are the deviations from their respective means.

$\therefore$  If  $X_i, Y_i$  are elements of  $x$  and  $y$  series respectively, then  $X_i - \bar{x} = x_i$  and  $Y_i - \bar{y} = y_i$

$$\therefore \sum x_i y_i = \sum (X_i - \bar{x})(Y_i - \bar{y}) = 60, \sum x_i^2 = \sum (X_i - \bar{x})^2 = 90$$

$$\text{Now, } \sigma_x^2 = \frac{\sum (X_i - \bar{x})^2}{n}$$

$$\therefore \sigma_x^2 = \frac{90}{n}$$

$$\therefore \sigma_x = \sqrt{\frac{90}{n}}$$

$$\text{Also, } \text{Cov}(X, Y) = \frac{1}{n} \sum (X_i - \bar{x})(Y_i - \bar{y})$$

$$\therefore \text{Cov}(X, Y) = \frac{60}{n}$$

$$r = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$$

$$\therefore 0.8 = \frac{\frac{60}{n}}{\sqrt{\frac{90}{n}} \times 2.5}$$

$$\therefore 0.8 \times 2.5 \times \sqrt{\frac{90}{n}} = \frac{60}{n}$$

$$\therefore 2 \times \frac{\sqrt{90}}{\sqrt{n}} = \frac{60}{n}$$

$$\therefore \frac{n}{\sqrt{n}} = \frac{60}{2 \times \sqrt{90}}$$

$$\therefore \frac{\sqrt{n} \times \sqrt{n}}{\sqrt{n}} = \frac{30}{\sqrt{90}} = \frac{\sqrt{30} \times \sqrt{30}}{\sqrt{3}\sqrt{30}}$$

$$\therefore \sqrt{n} = \sqrt{10}$$

$$\therefore n = 10$$

### Miscellaneous Exercise 5 | Q 5 | Page 64

A sample of 5 items is taken from the production of a firm. Length and weight of 5 items are given below. [Given :

$$\sqrt{0.8823} = 0.9393]$$

|                     |   |    |    |    |    |
|---------------------|---|----|----|----|----|
| <b>Length (cm)</b>  | 3 | 4  | 6  | 7  | 10 |
| <b>Weight (gm.)</b> | 9 | 11 | 14 | 15 | 16 |

Calculate the correlation coefficient between length and weight and interpret the result.

#### **SOLUTION**

Let length =  $x_i$  (in cm), Weight =  $y_i$  (in gm)

|              | $x_i$     | $y_i$     | $x_i^2$    | $y_i^2$    | $x_i y_i$  |
|--------------|-----------|-----------|------------|------------|------------|
|              | 3         | 9         | 9          | 81         | 27         |
|              | 4         | 11        | 16         | 121        | 44         |
|              | 6         | 14        | 36         | 196        | 84         |
|              | 7         | 15        | 49         | 225        | 105        |
|              | 10        | 16        | 100        | 256        | 160        |
| <b>Total</b> | <b>30</b> | <b>65</b> | <b>210</b> | <b>879</b> | <b>420</b> |

From the table, we have

$$n = 5, \sum x_i = 30, \sum y_i = 65, \sum x_i^2 = 210, \sum y_i^2 = 879, \sum x_i y_i = 420$$

$$\bar{x} = \frac{\sum x_i}{n} = \frac{30}{5} = 6$$

$$\bar{y} = \frac{\sum y_i}{n} = \frac{65}{5} = 13$$

$$\text{Cov}(X, Y) = \frac{1}{n} \sum x_i y_i - \bar{x} \bar{y}$$

$$= \frac{1}{5} \times 420 - 6 \times 13$$

$$= 84 - 78$$

$$= 6$$

$$\sigma_x^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$= \frac{210}{5} - (6)^2$$

$$= 42 - 36$$

$$\therefore \sigma_x^2 = 6$$

$$\therefore \sigma_x = \sqrt{6}$$

$$\sigma_y^2 = \frac{\sum y_i^2}{n} - (\bar{y})^2$$

$$= \frac{879}{5} - (13)^2$$

$$= 175.8 - 169$$

$$\sigma_y^2 = 6.8$$

$$\therefore \sigma_y = \sqrt{6.8}$$

Thus, the coefficient of correlation between X and Y is

$$\begin{aligned}
 r &= \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y} \\
 &= \frac{6}{\sqrt{6}\sqrt{6.8}} \\
 &= \sqrt{\frac{6}{6.8}} \\
 &= \sqrt{\frac{60}{68}} \\
 &= \sqrt{\frac{15}{17}} \\
 &= \sqrt{0.8823}
 \end{aligned}$$

$$\therefore r = 0.9393 \approx 0.94$$

$\therefore$  The value of  $r$  indicates high degree of positive correlation between length and weight of items.

### Miscellaneous Exercise 5 | Q 6 | Page 64

Calculate the correlation coefficient from the following data, and interpret it.

|          |    |    |   |   |   |    |    |
|----------|----|----|---|---|---|----|----|
| <b>X</b> | 1  | 3  | 5 | 7 | 9 | 11 | 13 |
| <b>Y</b> | 12 | 10 | 8 | 6 | 4 | 2  | 0  |

### **SOLUTION**

|  | $x_i$ | $y_i$ | $x_i^2$ | $y_i^2$ | $x_i y_i$ |
|--|-------|-------|---------|---------|-----------|
|  | 1     | 12    | 1       | 144     | 12        |
|  | 3     | 10    | 9       | 100     | 30        |
|  | 5     | 8     | 25      | 64      | 40        |
|  | 7     | 6     | 49      | 36      | 42        |
|  | 9     | 4     | 81      | 16      | 36        |

|              |           |           |            |            |            |
|--------------|-----------|-----------|------------|------------|------------|
|              | 11        | 2         | 121        | 4          | 22         |
|              | 13        | 0         | 169        | 0          | 0          |
| <b>Total</b> | <b>49</b> | <b>42</b> | <b>455</b> | <b>364</b> | <b>182</b> |

From the table, we have

$$n = 7, \sum x_i = 49, \sum y_i = 42, \sum x_i^2 = 455, \sum y_i^2 = 364, \sum x_i y_i = 182$$

$$\therefore \bar{x} = \frac{\sum x_i}{n} = \frac{49}{7} = 7,$$

$$\bar{y} = \frac{\sum y_i}{n} = \frac{42}{7} = 6$$

$$\text{Cov}(X, Y) = \frac{1}{n} \sum x_i y_i - \bar{x} \bar{y}$$

$$= \frac{1}{7} \times 182 - (7 \times 6)$$

$$= 26 - 42$$

$$\therefore \text{Cov}(X, Y) = -16$$

$$\sigma_x^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$= \frac{455}{7} - (7)^2$$

$$= 65 - 49$$

$$\therefore \sigma_x^2 = 16$$

$$\therefore \sigma_x = 4$$

$$\sigma_y^2 = \frac{\sum y_i^2}{n} - (\bar{y})^2$$

$$= \frac{364}{7} - (6)^2$$

$$= 52 - 36$$

$$\sigma_y^2 = 16$$

$$\therefore \sigma_y = 4$$

Thus, the coefficient of correlation between X and Y is

$$r = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y}$$

$$= \frac{-16}{4 \times 4}$$

$$= -1$$

$\therefore$  The value of r indicates perfect negative correlation between x and y.

#### Miscellaneous Exercise 5 | Q 7 | Page 64

Calculate the correlation coefficient from the following data and interpret it.

|          |    |    |    |    |    |    |    |
|----------|----|----|----|----|----|----|----|
| <b>x</b> | 9  | 7  | 6  | 8  | 9  | 6  | 7  |
| <b>y</b> | 19 | 17 | 16 | 18 | 19 | 16 | 17 |

#### **SOLUTION**

|  | <b>x<sub>i</sub></b> | <b>y<sub>i</sub></b> | <b>x<sub>i</sub><sup>2</sup></b> | <b>y<sub>i</sub><sup>2</sup></b> | <b>x<sub>i</sub>y<sub>i</sub></b> |
|--|----------------------|----------------------|----------------------------------|----------------------------------|-----------------------------------|
|  | 9                    | 19                   | 81                               | 361                              | 171                               |
|  | 7                    | 17                   | 49                               | 289                              | 119                               |
|  | 6                    | 16                   | 36                               | 256                              | 96                                |
|  | 8                    | 18                   | 64                               | 324                              | 144                               |
|  | 9                    | 19                   | 81                               | 361                              | 171                               |

|              |           |            |            |             |            |
|--------------|-----------|------------|------------|-------------|------------|
|              | 6         | 16         | 36         | 256         | 96         |
|              | 7         | 17         | 49         | 289         | 119        |
| <b>Total</b> | <b>52</b> | <b>122</b> | <b>396</b> | <b>2136</b> | <b>916</b> |

From the table, we have

$$n = 7, \sum x_i = 52, \sum y_i = 122, \sum x_i^2 = 396, \sum y_i^2 = 2136, \sum x_i y_i = 916$$

$$\therefore \bar{x} = \frac{\sum x_i}{n} = \frac{52}{7}$$

$$\bar{y} = \frac{\sum y_i}{n} = \frac{122}{7}$$

$$\therefore \bar{x}\bar{y} = \frac{52 \times 122}{49} = \frac{6344}{49}$$

$$\text{Cov}(X, Y) = \frac{1}{n} \sum x_i y_i - \bar{x}\bar{y}$$

$$= \frac{916}{7} - \frac{6344}{49}$$

$$= \frac{6412 - 6344}{49}$$

$$= \frac{68}{49}$$

$$\sigma_x^2 = \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$= \frac{396}{7} - \left(\frac{52}{7}\right)^2$$

$$\begin{aligned}
&= \frac{8772 - 2704}{49} \\
&= \frac{68}{49} \\
\sigma_y^2 &= \frac{\sum y_i^2}{n} - (\bar{y})^2 \\
&= \frac{2136}{7} - \left(\frac{122}{7}\right)^2 \\
&= \frac{14952 - 14884}{49} \\
&= \frac{68}{49} \\
\therefore \sigma_x \sigma_y &= \sqrt{\sigma_x^2 \sigma_y^2} \\
&= \sqrt{\frac{68}{49} \times \frac{68}{49}} \\
&= \frac{68}{49} \\
r &= \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y} \\
&= \frac{\left(\frac{68}{49}\right)}{\frac{68}{49}} = 1
\end{aligned}$$

$\therefore$  The value of  $r$  indicates perfect positive correlation between  $x$  and  $y$ .

#### Miscellaneous Exercise 5 | Q 8. (i) | Page 64

If the correlation coefficient between  $x$  and  $y$  is 0.8, what is the correlation coefficient between  $2x$  and  $y$

### **SOLUTION**

Correlation coefficient remains unaffected by the change of origin and scale.

i.e., if  $u_i = \frac{x_i - a}{h}$  and  $v_i = \frac{y_i - b}{k}$ , then  $\text{Corr}(u, v) = \pm \text{Corr}(x, y)$ , according to the same or opposite signs of  $h$  and  $k$ .

$$u_i = \frac{2(x_i - 0)}{1}, v_i = \frac{y_i - 0}{1}$$

Here,  $h = 1$  and  $k = 1$  are of the same signs.

$$\therefore \text{Corr}(u, v) = \text{Corr}(x, y) = 0.8$$

### **Miscellaneous Exercise 5 | Q 8. (ii) | Page 64**

If the correlation coefficient between  $x$  and  $y$  is 0.8, what is the correlation coefficient between  $X/2$  and  $y$

### **SOLUTION**

Correlation coefficient remains unaffected by the change of origin and scale.

i.e., if  $u_i = \frac{x_i - a}{h}$  and  $v_i = \frac{y_i - b}{k}$ , then  $\text{Corr}(u, v) = \pm \text{Corr}(x, y)$ , according to the same or opposite signs of  $h$  and  $k$ .

$$u_i = \frac{(x_i - 0)}{2}, v_i = \frac{y_i - 0}{1}$$

Here,  $h = 1$  and  $k = 1$  are of the same signs.

$$\therefore \text{Corr}(u, v) = \text{Corr}(x, y) = 0.8$$

### **Miscellaneous Exercise 5 | Q 8. (iii) | Page 64**

If the correlation coefficient between  $x$  and  $y$  is 0.8, what is the correlation coefficient between  $x$  and  $3y$

### **SOLUTION**

Correlation coefficient remains unaffected by the change of origin and scale.

i.e., if  $u_i = \frac{x_i - a}{h}$  and  $v_i = \frac{y_i - b}{k}$ , then  $\text{Corr}(u, v) = \pm \text{Corr}(x, y)$ , according to the same or opposite signs of  $h$  and  $k$ .

$$\text{Corr}(x, 3y) = \text{Corr}(x, y) = 0.8$$

### **Miscellaneous Exercise 5 | Q 8. (iv) | Page 64**

If the correlation coefficient between  $x$  and  $y$  is 0.8, what is the correlation coefficient between  $x - 5$  and  $y - 3$

### **SOLUTION**

Correlation coefficient remains unaffected by the change of origin and scale.

i.e., if  $u_i = \frac{x_i - a}{h}$  and  $v_i = \frac{y_i - b}{k}$ , then  $\text{Corr}(u, v) = \pm \text{Corr}(x, y)$ , according to the same or opposite signs of  $h$  and  $k$ .

$$\text{Corr}(x - 5, y - 3) = \text{Corr}(x, y) = 0.8$$

### **Miscellaneous Exercise 5 | Q 8. (v) | Page 64**

If the correlation coefficient between  $x$  and  $y$  is 0.8, what is the correlation coefficient between  $x + 7$  and  $y + 9$

### **SOLUTION**

Correlation coefficient remains unaffected by the change of origin and scale.

i.e., if  $u_i = \frac{x_i - a}{h}$  and  $v_i = \frac{y_i - b}{k}$ , then  $\text{Corr}(u, v) = \pm \text{Corr}(x, y)$ , according to the same or opposite signs of  $h$  and  $k$ .

$$\text{Corr}(x + 7, y + 9) = \text{Corr}(x, y) = 0.8$$

### **Miscellaneous Exercise 5 | Q 8. (vi) | Page 64**

If the correlation coefficient between  $x$  and  $y$  is 0.8, what is the correlation coefficient

between  $\frac{x - 5}{7}$  and  $\frac{y - 3}{8}$ ?

### **SOLUTION**

Correlation coefficient remains unaffected by the change of origin and scale.

i.e., if  $u_i = \frac{x_i - a}{h}$  and  $v_i = \frac{y_i - b}{k}$ , then  $\text{Corr}(u, v) = \pm \text{Corr}(x, y)$ , according to the same or opposite signs of  $h$  and  $k$ .

$$\text{Corr}\left(\frac{x - 5}{7}, \frac{y - 3}{8}\right) = \text{Corr}(x, y) = 0.8$$

### **Miscellaneous Exercise 5 | Q 9 | Page 64**

In the calculation of the correlation coefficient between the height and weight of a group of students of a college, one investigator took the measurements in inches and pounds while the other investigator took the measurements in cm. and kg. Will they get the same value of the correlation coefficient or different values? Justify your answer.

### **SOLUTION**

Coefficient of correlation is a ratio of covariance and standard deviations.

Since covariance and standard deviations are independent of units of measurement.

∴ Coefficient of correlation is also independent of units of measurement.

∴ Values of the coefficient of correlation obtained by first and second investigators are the same.